Yuanqiang Yu

+86 176 9595 0835 yuyuanqiang@gmail.com Google Scholar

Education

Tianjin University (QS 235) **Sep 2020 – Mar 2023**

M.Sc. in Computer Science and Technology; ***GPA: 87/100*** *Tianjin, China*

- Advisor: Prof. Jianye Hao.
- Research Focus: Deep Reinforcement Learning, Multitask Reinforcement Learning, and Transfer Learning.
- Relevant Coursework: Advanced Algorithms (94), Image Synthesis (92), Natural Language Processing (91), Deep Learning (90), Comprehensive Experiments on Big Data (89), and Stochastic Processes (87).

Tiangong University **Sep 2016 – Jun 2020**

B.Eng. in Computer Science and Technology; ***GPA: 3.82/4.0 (top 1%)*** *Tianjin, China*

- Relevant Coursework: Machine Learning (97), Curriculum Design of Algorithm and Data Structure (100), Linear Algebra (98), Computer Networks (96), Discrete Mathematics (93), and Probability Theory and Mathematical Statistics (92).

Research Interests

My research focuses on reinforcement learning (RL) and large language model (LLM) agents. My earlier research focused on deep reinforcement learning, especially multitask reinforcement learning and transfer learning. I now work on LLM post-training and LLM agents, with an emphasis on agent memory systems and skill routing. I am also interested in self-evolving agents, context management, and long-horizon planning under compute and context constraints.

Work Experience

Alibaba Group, Taobao Marketplace **Oct 2025 – Present**

Machine Learning Engineer *Hangzhou, China*

- Developed a **Taobao merchant agent** built on **Qwen3-30B-A3B**, using multitask SFT and RL to improve long-horizon planning. Dynamically adjusted rollout allocation based on the learning progress of each task to improve sample efficiency; instruction following and answer accuracy improved by **15% on average** over the baseline on an internal evaluation set.
- Built a **two-tier agent memory system** to preserve user and task context across conversations, covering user profiles, memories of product operations, and experiences with skills and tools; combined asynchronous extraction, confidence-based lifecycle management, hybrid retrieval combining BM25 lexical search with dense semantic search, and LLM reranking to reduce **average turns per conversation by 13.6%** in online deployment versus the same agent without memory.
- Co-designed **SkillRouter** to make routing accurate and scalable across approximately **80,000 candidate skills**, using full skill bodies in a compact 1.2B-parameter two-stage retrieval and reranking pipeline; achieved **74.0% Hit@1** with **13× fewer parameters** and **5.8× faster inference** than the strongest evaluated baseline.

Huawei Technologies, 2012 Laboratories **Jun 2023 – Oct 2025**

Machine Learning Engineer *Shenzhen, China*

- Contributed to the **MindRLHF** project as a core developer, with responsibility for adapting **GRPO, DAPO, and GSPO** to **Ascend 910C** accelerators. Designed **zero-redundancy actor model resharding** between training and generation phases to reduce communication overhead, reducing resharding latency by **25%**; developed **adaptive sequence packing** to reduce padding overhead, increasing training throughput by **3–7×**.
- Developed shared memory support between the host and Ascend devices for the **Virtual Optimizer** in the **MindSpore** project (**9K+ stars**), reducing optimizer memory use in pipeline parallel training. Mapped optimizer states in host memory directly into the Ascend device address space to avoid optimizer swapping, achieving a **1.5× training speedup**.
- Built a **financial-risk assessment agent** covering **more than 5,000 suppliers**; extracted **over 200 features** from business documents and applied **Isolation Forest** to identify anomalous values across invoice fields. Fine-tuned a **38B-parameter LLM** with LoRA and used **RAG** to retrieve experience from earlier invoice anomalies detected by Isolation Forest and provide it as context for document-based risk assessment.

Publications

Deep Reinforcement Learning

[1] **Yuanqiang Yu**, Yanzhao Zheng, Zhentao Zhang, Tianze Xu, Chao Ma, Jihuai Zhu, Jiashun Liu, Xinle Deng, Baohua Dong, Hangcheng Zhu, and Ruohui Huang. "**PAC: Progress-Augmented Advantage Curriculum for Multi-Task Reinforcement Learning of LLMs**." *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2026.

[2] **Yuanqiang Yu**, Peng Zhang, Kai Zhao, Yan Zheng, and Jianye Hao. "**Accelerating Deep Reinforcement Learning via Knowledge-Guided Policy Network**." *Autonomous Agents and Multi-Agent Systems*, 37, Article 17, 2023.

[3] **Yuanqiang Yu**, Tianpei Yang, Yongliang Lv, Yan Zheng, and Jianye Hao. "**T3S: Improving Multi-Task Reinforcement Learning with Task-Specific Feature Selector and Scheduler**." *International Joint Conference on Neural Networks (IJCNN)*, pp. 1–8, 2023.

[4] Yongliang Lv, **Yuanqiang Yu**, Yan Zheng, Jianye Hao, Yongming Wen, and Yue Yu. "**Limited Information Opponent Modeling**." *International Conference on Artificial Neural Networks (ICANN)*, pp. 511–522, 2023.

[5] Min Wang, Xingzhong Wang, Wei Luo, Yixue Huang, and **Yuanqiang Yu**. "**Accelerating Deep Reinforcement Learning Under the Guidance of Adaptive Fuzzy Logic Rules**." *2022 Prognostics and Health Management Conference (PHM-2022 London)*, pp. 350–359, 2022.

LLM Agents

[6] Yanzhao Zheng, **Yuanqiang Yu**, Tianze Xu, Chao Ma, Zhentao Zhang, Jihuai Zhu, Baohua Dong, Hangcheng Zhu, and Ruohui Huang. "**Counterfactual Constraint-Conditioned On-Policy Distillation for Multi-Constraint Instruction Following**." *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2026.

[7] Yanzhao Zheng, Zhentao Zhang, Chao Ma, **Yuanqiang Yu**, Jihuai Zhu, Yong Wu, Tianze Xu, Baohua Dong, Hangcheng Zhu, Ruohui Huang, and Gang Yu. "**SkillRouter: Skill Routing for LLM Agents at Scale**." *Conference on Language Modeling (COLM)*, 2026.

[8] Yong Wu, Yanzhao Zheng, Tianze Xu, Zhentao Zhang, **Yuanqiang Yu**, Jihuai Zhu, Chao Ma, Binbin Lin, Baohua Dong, Hangcheng Zhu, Ruohui Huang, and Gang Yu. "**ContextBudget: Budget-Aware Context Management for Long-Horizon Search Agents**." *Conference on Language Modeling (COLM)*, 2026.

[9] Tianze Xu, Yanzhao Zheng, Pengrui Lu, Lyumanshan Ye, Yong Wu, Zhentao Zhang, **Yuanqiang Yu**, Chao Ma, Jihuai Zhu, Pengfei Liu, Baohua Dong, Hangcheng Zhu, Ruohui Huang, and Gang Yu. "**Rubrics to Tokens: Bridging Response-level Rubrics and Token-level Rewards in Instruction Following Tasks**." *Conference on Language Modeling (COLM)*, 2026.

Selected Honors and Awards

Outstanding Winner (Top 1%), 2018 Interdisciplinary Contest in Modeling (ICM)	2018
First Prize (Top 2%), "Huawei Cup" The 18th China Post-Graduate Mathematical Contest in Modeling	2021
First Prize (Top 10%), Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM)	2019
Second Prize, The 7th "Teddy Cup" Data Mining Challenge	2019
Second Prize, 2018 Asia and Pacific Mathematical Contest in Modeling (APMCM)	2018
Second Prize, 2017 National English Competition for College Students (NECCS)	2017
Tianjin Municipal Government Scholarship (Top 1%)	2018

Skills

Programming: Python, C/C++, Java, Go, JavaScript, TypeScript
Languages: English (Advanced), Mandarin (Native)
LLM Agents: Skill Routing, Agent Memory, Context Management, RAG
LLM Post-Training and RL: SFT, LoRA, PPO, GRPO, DAPO, GSPO
Frameworks: PyTorch, MindSpore, CUDA, scikit-learn, FastAPI, Redis
Systems: Linux, Shell, Git, Docker